PROJECT-BASED LEARNING

PHASE-I EVALUATION

Cloudbalance: Intelligent load balancing and resource allocation

Team ID: T140

Department of Computer Science & Engineering
Graphic Era (Deemed to be University), Dehradun
Academic Session 2026–27

Team & Mentor Details

Team Information

Team ID: T140
Semester: 3rd
Section: J,DS-2,DS-1,Cyber
Domain: *DS+CPP*

Team Members

1. *Amrit Raj – 2510010756*
2. *Pawani Gupta – 2510380107*
3. *Anvi Shandilya – 2510380107*
4. *Rushil Kumar – 2510350004*

Mentor

Mr.Piyush Agarwal **Interactions completed:** ______

Problem Statement & Motivation

Problem Statement

Cloud environments handle multiple tasks across multiple servers. Uneven workload distribution can overload some servers while leaving others underutilized, resulting in higher response time and inefficient resource usage. CloudBalance aims to intelligently distribute incoming tasks among available resources based on their current workload.

Motivation

- *Uneven workload distribution among cloud servers..*
- *Overloaded servers can reduce system performance.*
- *Simple methods may not handle*

Target Users / Stakeholders

- *Cloud administrators*
- *Cloud server users*
- *Better resource utilizations*

Objectives

Project Objectives

1. *Devlop a system to balance workloads along diffrent servers.*
2. *Allocate task based on server load and resource avaibility.*
3. *Integrate DSA in C and OOp with C++ cpncepts.*
4. *Test and evaluate the system using suitable test case.*
5. *Reduce server load.*

Key Objective: *State the main measurable outcome of the project.*

Proposed Solution

Proposed Approach

1. *Collect server details*
2. *Analyze server load and select best server*
3. *Assign tasks according to server availability*
4. *Stores server allocation details*
5. *Display server load details*

Key Features

- *Load based task allocation*
- *Efficient resource utilization*
- *Server load monitoring*
- *Dynamic task distribution*
- *Simple result visualization*

Clearly explain how the proposed solution addresses the identified problem.

Integration of PBL Course Concepts

Course	Concepts Applied	Project Module
Data Structures in C	*Arrays, Linked Lists, Sorting.*	*DSA with C*
OOPs with C++	*Classes, Inheritance, Polymorphism.*	*OOP using C++*

Requirement: Demonstrate meaningful integration of both subjects in **one** project.

System Architecture / Workflow

INPUT → PROCESSING → LOAD BALANCE → MANAGEMENT → OUTPUT

Major Components

- *Server Task Input Module*
- *Load Balancing Module*
- *Resource Module*

Data / Control Flow

- *User provides task and server details*
- *System Analyse Server Workload*
- *Task is allocated and result is displayed*

Technology Stack & Methodology

Technology Stack

- Programming: *C / C++*
- Database: *Not Required*
- Tools / Framework: *Std C/C++ Libraries*
- IDE: *Visual Studio Code*
- Version Control: *Git / GitHub*

Development Methodology

1. Requirement Analysis
2. System Design
3. Implementation
4. Testing
5. Evaluation

Justify the selection of technologies and methodology.

Initial Progress & Team Contribution

Progress Achieved

- *Background / literature study*
- *Requirements identified*
- *Architecture prepared*
- *Initial module implemented*
- *Database / data preparation*

Individual Contribution

Member	Role	Contribution
M1	*Lead*	*Project Planning*
M2	*Developer*	*Load Balancer Devl-opment*
M3	*DB/Testing*	*Testing*
M4	*Developer/Docs*	*Presentation*

Roadmap, Expected Outcomes & References

Roadmap

1. Phase-I: Proposal & Design
2. Phase-II: Development
3. Phase-III: Final Implementation

Expected Outcomes

- *Working prototype / system*
- *Measurable results*
- *Practical applicability*
- *Future scalability*

References

1. *Cloud Load Balancing - IEEE Xplore*
2. *Cloud Computing - Thomas erl*
3. *AWS Elastic Load Balancing*
4. *GitHub - Cloud Load Balancer*

Thank You
Questions & Suggestions